

1 One method of testing a person's reaction time is to use an online, computer click timer test. Some students carried out a test using the method described.

- Each student looks at a computer screen and clicks a **start** button on the screen and waits for the background colour to change.
- As soon as the background colour changes the student clicks a **stop** button on the screen.
- The reaction time appears on the screen.

A student in the class thought that boys had a faster reaction time than girls.

A group of students from the class decided to repeat the test to investigate the hypothesis:

Boys have a faster reaction time than girls.

(a) (i) Identify the independent variable **and** the dependent variable in this investigation.

independent

dependent [2]

(ii) State the type of qualitative variable which best describes the independent variable.

..... [1]

- (iv) Suggest **one** feature of the online test which may be a source of error.

.....
..... [1]

Another group of students decided to test the hypothesis:

Right-handed people have a faster reaction time with their right hand than with their left hand.

Table 1.1 shows their results.

Table 1.1

student	reaction time using right hand/s	reaction time using left hand/s
A	0.26	0.28
B	0.23	0.23
C	0.22	0.25
D	0.25	0.22
E	0.26	0.27
F	0.24	0.24
G	0.28	0.26
H	0.25	0.30
I	0.21	0.25
J	0.26	0.29
K	0.27	0.31
mean	0.25	0.26
mode	0.26	
median		0.26

- (b) Complete Table 1.1 to show the mode and median values. [1]

- (c) (i) The students decided to carry out a *t*-test to find out if the difference in reaction time was significant.

State why a *t*-test is suitable for these data.

.....
..... [1]

(ii) State a null hypothesis for this test.

.....
.....
.....
..... [1]

(iii) State the number of degrees of freedom for this t -test.

.....
..... [1]

Table 1.2 shows the probability values of t .

Table 1.2

degrees of freedom	10	12	14	16	18	20	22	24	26	28	30	40	50	60
probability 0.05	2.23	2.18	2.14	2.12	2.10	2.09	2.07	2.06	2.06	2.05	2.04	2.02	2.01	2.00

(iv) The students concluded that the difference in reaction time was **not** significantly different.

State what this tells you about their calculated value of t .

.....
..... [1]

One student noticed that the more repetitions of the online test that were carried out, the faster the reaction time. The student decided to carry out a different test to investigate the effect of repetition and feedback on the accuracy of carrying out a task.

Three students each carried out a task which involved drawing straight lines, without using a ruler, on sheets of paper.

The three students did the task in separate rooms.

- Each student drew 20 straight lines, as close as possible to 7.5 cm long, on separate pieces of paper.
- The teacher measured all the lines and recorded the lengths to the nearest 0.1 cm in a table.
- The teacher gave feedback to each student by telling them the actual length of line 20.
- Each student then drew another 20 lines which were measured by the teacher and recorded in the table.

Table 1.3 shows the data that the teacher recorded.

Table 1.3

line	length of measured line before feedback / cm			length of measured line after feedback / cm		
	student 1	student 2	student 3	student 1	student 2	student 3
1	7.2	5.0	7.3	7.9	7.5	5.2
2	7.5	4.3	7.4	7.0	7.2	5.2
3	7.4	4.0	7.7	7.5	7.7	5.2
4	7.5	3.9	7.9	7.3	7.6	4.9
5	8.0	4.2	8.2	7.0	7.3	6.3
6	7.5	4.3	8.8	7.2	7.7	6.5
7	7.5	4.0	8.8	7.0	7.4	6.3
8	7.2	4.0	9.5	7.2	8.0	6.2
9	7.1	4.4	9.3	7.0	7.6	6.2
10	7.4	3.8	9.0	7.1	7.2	5.7
11	7.2	7.9	9.3	7.0	7.5	6.7
12	7.4	4.0	8.8	7.0	6.8	6.6
13	7.5	4.2	9.7	7.6	7.2	6.6
14	7.5	4.0	9.7	8.0	7.6	6.6
15	7.4	4.2	10.0	8.0	7.4	6.4
16	6.5	4.1	10.0	8.2	7.4	6.6
17	6.7	4.4	9.6	7.9	7.0	6.5
18	6.8	4.0	9.8	8.5	7.5	6.2
19	7.9	4.2	9.7	7.8	7.7	6.9
20	7.5	4.0	10.2	7.5	7.6	6.1
mean ± s	7.3 ± 0.36	4.2 ± 0.26	9.0 ± 0.90	7.5 ± 0.47	7.4 ± 0.27	6.1 ± 0.59

(d) When one of the students was drawing one of their lines, a person entered the room and distracted them. It was decided this result was anomalous.

(i) Indicate, by drawing a circle on Table 1.3, the length which is an anomalous result. [1]

(ii) Suggest what the students should do with the anomalous result.

.....

..... [1]

(iii) Identify the student who drew lines that were most **inconsistent** in length **and** give a reason for your choice.

student

reason [1]

(iv) The student concluded that receiving feedback on performance leads to an improvement in accuracy.

State **one** way in which the data support the conclusion **and** one way in which it does not.

supports

.....

.....

does not support

.....

.....

[2]

[Total: 20]